Varun Totakura

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Professional Summary

Passionate AI/ML Engineer with over 5 years of professional experience. Excellent team player, individual contributor and good leader with strong interpersonal skills, committed to achieving project objectives efficiently.

Technical Skills

- Frameworks: LangGraph, PyTorch, TensorFlow, Scikit-Learn, Keras, LangChain, Hugging Face
- Languages: Python, SQL, Javascript
- Modules: Dask, Numpy, Pandas, PySpark, Streamlit, FastAPI, Claude, OpenAI, Gemini, Llama
- ML Clouds: VertexAI, Databricks, Jenkins, Docker, Airflow, Kafka, AutoML, Azure AI
- Databases: BigQuery, SnowFlake, PostgreSQL, MySQL, Teradata, MongoDB, Cassandra, Pinecone, Chroma
- Clouds: Google Cloud, Microsoft Azure, Amazon Web Services, ServiceNow
- Visualization and Reports: Matplotlib, Seaborn, Plotly, Power BI, Tableau

Professional Experience

AI Engineer - Lead Assistant Manager, EXL Service Inc., New York, NY (Remote) *October 2024 – Present*

- Leading development of a LangChain-based multi-agent analytics platform with specialized agents for A/B testing, causal inference, and call analytics, backed by data validation checks and automated regression tests, guiding leadership decisions on healthcare marketing campaigns reaching 5M members and contributing to $1.5M in annual revenue.
- Leading a team of 5 data scientists by translating business goals into sprint tasks, assigning responsibilities, coordinating development, and consolidating deliverables to ensure timely project execution.
- Build a production RAG assistant with LangGraph and Pinecone, using [hybrid retrieval with reranking] over policy documents and a knowledge graph, with validation, logging, and monitoring; apply LLM to call transcripts to uncover frequent-caller pain points and anticipate their next questions.
- Established an LLM evaluation framework (Recall@K, groundedness, hallucination rate, latency, cost per query) on a [N]-question test set, raising Recall@5 from 68% to 81%, halving hallucinations from 12% to 6%, and cutting query cost 30% through better prompts, chunking, metadata retrieval, and knowledge-graph enrichment.
- Optimized the data pipeline behind the analytics dashboard by applying sparse data techniques, processing 7M+ records and cutting runtime and rendering time from 20 minutes to under 30 seconds (40x faster), enabling near real-time insights for leadership.
- Driving healthcare marketing campaigns through End-to-End ML solutions and A/B testing, incorporating model versioning, scalability, bias mitigation, and drift detection to enhance user experience, increase product sales, and target $1M in cost savings.
- Developed a noval approach in machine learning model using *Group Classification for Deep Imbalance Regression* to address highly skewed data, boosting performance by 3x times.
- Worked with diverse healthcare data sources, including claims, SDOH, medical notes, membership records, and chronic disease having 100+ million records, efficiently queried and processed using SnowFlake and Teradata.
- Leveraged Claude to extract features from medical notes, integrated them with tabular data, and achieved a 10% performance improvement.
- Engineered a multi-class classification model, elevating performance by nearly 2x times through rigours optimization and bias mitigation strategies.
- Engineered 15 CI/CD ML pipelines deployment using Jenkins and Apache Airflow DAGs on GCP, launching DataProc clusters to process data and generate predictions, with results stored in BigQuery tables for downstream process.
- Optimized text processing code by parallelizing tasks with Dask and Multi-threading, reducing execution time from 4 hours to under 2 hours while maintaining accuracy.

AI/ML Engineer - Graduate Research Assistant, Florida State University, Tallahassee, FL *August 2022 – August 2024*

- Designed a groundbreaking hybrid AI architecture combining TCN, CNN, and GRU to analyze Alzheimer's Disease patterns, achieving a 35% performance boost through domain disparity reduction.
- Developed a state-of-the-art Active Learning Algorithm for Imperfect Oracles with LLMs, optimizing performance through parallel computing with Dask to achieve 4x speedup in model training.
- Pioneered a scalable deep learning pipeline for model correctness across prediction, processing over 300,000 records of 10 diverse datasets with high accuracy.

- Implemented an NLP solution using BERT and GAN networks for precise bug localization in code, analyzing 150,000+ social media posts with high precision.

AI Software Developer - Systems Engineer, Tata Consultancy Services, Hyderabad, India *July 2020 – September 2022*

- Developed a cutting-edge Customer Service Chatbot using Azure Bot Framework and Python, deployed on Microsoft Teams to interact with ServiceNow via REST API; reducing operational times by 40% and enhancing user satisfaction.
- Spearheaded the creation of AI-powered chatbots with Virtual Agent, integrating 9+ data sources to transform user engagement, achieving a 50% increase in interaction efficiency.
- Orchestrated seamless integrations between ServiceNow and 5 external platforms using REST and SOAP protocols in an Agile SDLC environment.
- Engineered scalable ServiceNow automations, leveraging intelligent process optimization to boost productivity by 45% across critical workflows.
- Led a high-stakes data migration project, leveraging AI-driven integrations to transfer 15 million records with unprecedented efficiency and reducing manual effort by 70%.
- Architected and maintained 15+ robust applications, incorporating advanced workflows, Service Portal, ACLs, and AI-enhanced scheduled jobs.
- Achieved 95% accuracy in cross-platform data synchronization, ensuring flawless information flow across multiple ServiceNow instances and external systems.
- Pioneered R&D initiatives by designing ML-driven automation solutions with AI Studio, amplifying task efficiency by an impressive 20x.

Research Publications

- LLIE-CvT: A Convolutional Vision Transformer for Low Light Image Enhancement. *Expected 2026*
- Active Learning with Imperfect Oracles using Text Data. *IJCNN 2026*
- MetaErr: Predicting Error Patterns in Deep Neural Networks using the Meta Model Data. *ICSM 2025*
- Master's Thesis - A Study on Deep Learning Models for Real World Applications. *August 2024*
- Prediction of Transmittable Diseases Rate in a Location Using ARIMA and GARCH. *November 2021*
- Improved Safety of Self-Driving Car using Voice Recognition through CNN. *October 2021*
- Prediction of Stock Trend for Swing Trades using Long Short-Term Memory Neural Network Model. *March 2020*
- Prediction of Animal Vocal Emotions using Convolutional Neural Network. *February 2020*
- An Integrated Approach to Sentiment Analysis using Machine Learning Algorithms. *April 2020*
- Published another 5 research papers and survey papers. *2019 - 2020*

Awards and Achievements

- Appreciation for peer-reviewing 5 papers for IEEE SMC 2026. *October 2026*
- Awarded for reviewing 10 papers for "Machine Learning with Applications" an Elsevier Journal. *October 2026*
- Three times Star of the Month, Tata Consultancy Services. *2021 - 2022*
- Five times On the Spot Awards, Tata Consultancy Services. *2021 - 2022*
- Learning Achievement Award, Tata Consultancy Services. *2021 - 2022*
- Ranked 2nd Topper in Academics, Guru Nanak Institutions. *2018 - 2019*
- Peer Reviewed research papers - ORCID ID - https://orcid.org/0000-0002-5114-5205 *2019 - 2026*
- Google Scholar - https://scholar.google.com/citations?user=ZbVUfuYAAAAJ *2019 - 2026*

Certifications and Trainings

- Claude Certified Architect - Anthropic *October 2026*
- Completed Deep Learning Specialization taught by Andrew Ng and other instructors. *2020*
- Cyber Security Internship at Hyderabad City Police, trained on spam email detection, and more. *October 2019*
- Ethical Hacking, IOT, and Web Development training by Imparta. *June 2019*

Education

Florida State University, Tallahassee, Florida, United States
Master of Science in Computer Science (Thesis) *August 2024*

Jawaharlal Nehru Technological University – Guru Nanak Institutions, Telangana, India
Bachelor of Technology in Computer Science & Engineering *September 2020*